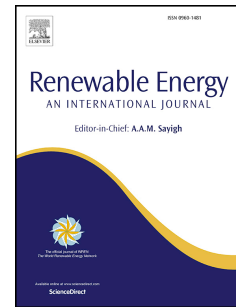


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Multi-Objective Optimal Sizing of Grid Connected Photovoltaic Batteryless System Minimizing the Total Life Cycle Cost and the Grid Energy

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Abstract

The increasing demand of electricity has led to drastic use of renewable energy resources because of the scarcity and environmental effects of fossil fuel. Photovoltaic energy is one of renewable energy sources that emerges exponentially due to its environmentally friendly, operating and maintenance free, among other benefits. However, the main drawback of photovoltaic energy is its unreliability due to unpredictable nature of solar. Thus, this paper tackles a novel optimization technique based on multi-objective sizing of grid connected photovoltaic without energy storage systems. The objective aims at minimizing the total life cycle cost (TLCC) and energy purchased from utility grid while maximizing the reliability. In this study, the microgrid system reliability is expressed by the loss of power supply probability (LPSP). Mixed integer linear programming has been used to determine the decision variables which are the optimal number of photovoltaic panels and the hourly powers from utility grid. A case study has been done based on daily power demand of Engineering workshops at Jomo Kenyatta University of Agriculture and Technology (JKUAT). From the results obtained, the optimal number is 354 photovoltaic arrays, 0.6022 kWh is purchased from the grid from 7:00 hrs to 18:00 hrs, and the total life cycle cost of the project is found to be 191630\$. The daily potential energy saving is up to 64.16%.

Keywords: Islanding mode, Levelized cost of energy, Loss of power supply probability, Mixed integer linear programming, Stand-alone.

1. Introduction

Owing to the increase in demand of electricity and growing environmental concerns, focus has shifted to generate power from renewable energy sources such as photovoltaic, wind, biomass, micro hydraulic, and tidal which do not emit green house gases [1]. Among these renewable energy resources, photovoltaic solar has gained widespread popularity due to the fact that it is noiseless, readily available, and environmentally friendly, among other factors [2]. Photovoltaic

^{*}Fully documented templates are available in the elsarticle package on CTAN.

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energy can be used either in islanding mode in order to supply electricity to remote areas or grid connected mode.

In several countries, efforts have been made to supplement conventional power generation by grid connected photovoltaic energy systems [3]. They have shown great advantages, particularly in urban areas where the utility grid is readily available. Additionally, they can reduce stress on national grid [2, 4]. Grid connected photovoltaic systems can offer end-users benefits of lowering power purchased from grid and selling the excess photovoltaic energy to the main grid.

In Kenya, various factors such as industrialization, standards of living, and population growth are increasing electricity demand. According to [5, 6], Kenya electricity consumption is increasing at an averaging rate of 6% annually and electricity access stands at the rate of 56%. Under Kenya vision 2030, the country aims at enhancing and diversifying national power generation by identifying new power generation sources [7, 8]. In fact, Kenya is endowed with very abundant solar irradiation that still remains unexploited for alternative energy solutions. Given that Kenya lies along the Equator with an average solar radiation of 6.1 kWh/m²/day [9, 10]. Currently, photovoltaic power generation in Kenya is less than 1% relative to current available generating power sources [7]. With its high potential, the country can harness grid connected photovoltaic solar energy to meet its future energy demand and lower the stress on the national grid.

Despite the ever-increasing installed grid connected photovoltaic solar energy worldwide, the total life cost of the system is still exorbitant. The main factor leading to high life cycle cost in grid connected photovoltaic energy is the connection of energy storage systems in order to enhance the reliability of microgrid [1, 11, 12]. However, energy storage systems have a short lifetime from 3 to 5 years, present high investment cost, operating & maintenance, and replacement costs, and exhibit power losses due to their internal resistances [13, 14].

Many studies have been performed for sizing grid connected photovoltaic energy, but most of them lack the potential to address the multi-objective problem since they dealt with one particular objective while disregarding the other objectives. In order to make comparison to other electricity generation technologies, the other objectives that need to be considered includes the total life cycle cost, which presents the economic analysis throughout the life of the project and the levelized cost of energy [15, 16]. For instance, in [2], optimal sizing grid connected photovoltaic without storage system was developed using genetic algorithm such that the net present value as economic factor was used to determine the optimum size. In [17], a sizing study related to cost analysis of photovoltaic grid connected hybrid energy system in Morocco using sunny design software was examined. Hybrid optimization model for electric renewable software was used to minimize the net present cost and carry out an economical expedience [3, 18]. A single objective to minimize the difference between the electrical energy generation and energy demand on the basis of an iterative approach is examined in [19]. In [20], an optimal sizing of grid photovoltaic considering three types of cells technologies namely polycrystalline cell, monocrystalline cell and thin film cell to minimize the investment cost are discussed.

In addition, other studies have been carried out considering small scale facilities like residential areas with battery storage system. For instance, in [21], an optimal sizing of grid connected photovoltaic with battery system is investigated to minimize the total annual cost of electricity under a time of use tariff using genetic algorithm. In [22], an optimal sizing of lithium battery storage energy for grid connected photovoltaic systems in order to minimize the cost of stored energy is examined. In [23], an optimal design of grid connected photovoltaic system on rooftop area using analytic method was examined. In [24], an efficient algorithm to determine the optimal size of battery storage grid connected photovoltaic systems was introduced. The optimum size aimed at minimizing the cost associated with net power purchase from utility grid and battery capacity loss while satisfying the load demand. In [25], the optimal sizing of a battery storage grid connected energy system was presented. The optimal battery storage has been determined using direct search method based on hourly optimized energy flow for one year. In [26], an economic analysis of grid connected and stand-alone photovoltaic system was discussed on the basis of net present value. In [27], techno-economic analysis of grid connected photovoltaic system using photovoltaic geographical information system and photovoltaic watt's software was carried out based on the annual savings, environmental aspects and payback period.

Exorbitant costs related to energy storage units can curb the use grid connected photovoltaic system in large scale or industrial facilities. To reduce these costs, grid connected photovoltaic can be designed without energy storage system as pointed out in [28, 29]. As a result, reliability challenges can occur in grid connected photovoltaic systems when the storage energy systems are left out. Additionally, the multi-objective optimization was not within the focus of majority of previous work. Thus, this paper addresses a multi-objective optimization sizing in order to minimize the total life cycle cost and energy purchased from utility grid simultaneously whilst maximizing the reliability requirement of grid connected photovoltaic batteryless system. In this study, the reliability is expressed by the loss of power supply probability (LPSP). Designing such grid connected photovoltaic batteryless system that guarantees an effective cost and minimize the power purchased from the grid while satisfying reliability requirement is a challenging task. As the optimal number of photovoltaic arrays must be an integer figure and power from grid utility is a continuous value, mixed integer linear programming has been used to carry out the multi-objective optimal sizing. Additionally, several optimization methods based on mixed integer linear programming have been used for optimal sizing of hybrid photovoltaic grid connected and standalone applications [30, 31, 32]. The main advantages of mixed integer line programming approach over the heuristic algorithms are that the Linear Programming (LP) subproblems can be solved quickly and the linear constraints result in a convex feasible region which guarantee to obtain the global optimum [33].

This paper is organized as follows: section 2 exhibits the modelling of different components of grid connected photovoltaic system. In section 3 the economic analysis is presented. In section 4 the reliability analysis is given. In section 5 methodology description approach is discussed. In section 6 the case study is presented. In section 7 the results and discussions are shown. Finally conclusion drawn from the optimization design is summarized in section 8.

2. Modelling of grid connected photovoltaic system

The proposed grid connected photovoltaic energy system is composed of utility grid, photovoltaic plant, inverter DC/AC and various Alternating Current (AC) loads as illustrated in Fig. 1. As photovoltaic plant generates direct current (DC) power, it is connected to the point of common coupling (PCC) through the DC/AC power converter. Utility grid and loads are directly connected to the point of common coupling without any interfacing power conditioning device. The loads can be firstly supplied by photovoltaic solar power P_{pv} , while the utility grid P_{grid} acts as backup to provide load power demand whenever photovoltaic energy is insufficient. The proposed energy system does not necessitate dumping loads because the grid can receive surplus photovoltaic power generated. The grid supplies electricity when power grid is greater than zero ($P_{grid}(t) \geq 0$). The mathematical models of different components within the grid connected photovoltaic system are represented as follows

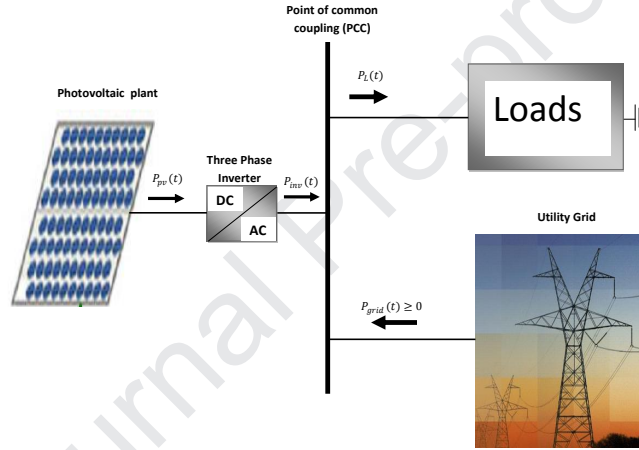


Figure 1: Schematic layout of the grid connected photovoltaic batteryless system

2.1. Photovoltaic energy System

Photovoltaic panels produce electricity from solar energy. A solar cell is the fundamental component to transform sunlight into electricity. The connection of many cells constitute a module whereas series-parallel connection of modules constitute a array [34, 35, 36, 37]. The power output of photovoltaic depends on the area of photovoltaic panel, efficiency of photovoltaic, solar radiance, and ambient temperature. The output power of a photovoltaic array is given by [36]

$$P_{pv}(t) = \eta_{pv} A_{pv} I(t), \quad (1)$$

where A_{pv} denotes the area of photovoltaic array, $I(t)$ denotes the solar irradiation incident on the photovoltaic array at time t , η_{pv} denotes the efficiency of the photovoltaic generator.

The simplified solar irradiation which is dependent on the time of the day is given by [29]

$$I(t) = R_B (I_B(t) + I_D(t)) + I_D(t), \quad (2)$$

where R_B is the geometric factor that represents the ratio of beam irradiation incident on the titled plane to the standard irradiation on the horizontal plane, $I_B(t)$ is the hourly global irradiation and

$I_D(t)$ is the diffuse irradiation.

Photovoltaic efficiency function of hourly insolation and ambient temperature is expressed as [36]

$$\eta_{pv} = \eta_r \eta_{pc} (1 - \beta(T_c - NOCT)), \quad (3)$$

where η_r denotes the manufacturer rated efficiency of the module making up the photovoltaic array, η_{pc} is the power conditioning efficiency, β is the generator efficiency temperature factor, ranging from 0.004-0.006 $^{\circ}C$, T_c cell temperature, and $NOCT$ denotes the normal operating cell temperature. A typical value of $NOCT$ equals to 45 $^{\circ}C$. The cell temperature is expressed as [36]

$$T_c = 30 + 0.0175(I(t) - 300) + 1.14(T_a - 25), \quad (4)$$

where T_a denotes the ambient temperature, while $I(t)$ is the solar irradiation at time t .

Additionally, in this study, the limited number of photovoltaic arrays, X_{pv} , is bounded such that,

$$N_{pv}^{min} \leq X_{pv} \leq N_{pv}^{max} \quad (5)$$

where N_{pv}^{min} is taken as zero, while $N_{pv}^{max} = \frac{A_T}{A_{pv}}$.

here A_T is the total surface area available at the site, and A_{pv} is the surface area of unit photovoltaic array.

2.2. Grid Energy

The grid is used as a backup power supply. The grid supplies power when photovoltaic power is not enough to meet the load power demand [18]. When grid power is greater than zero, electricity is purchased from the utility grid and when grid power is lesser than zero, electricity is sold to the utility grid. In this study, the grid can be within a specific range that is expressed as

$$0 \leq P_{grid}(t) \leq +\infty \quad (6)$$

here $P_{grid}(t)$ denotes the supplied power from the grid.

In this paper, the university setup where most of the activities take place between 7:00 hrs and 18:00 hrs is considered as a case study. The energy purchased from the main grid between 7:00 hrs and 18:00 hrs is given by

$$E_{grid} = \sum_{t=7}^{18} t_s P_{grid}(t) \quad (7)$$

where $P_{grid}(t)$ denotes the grid power, and t_s is the sampling time.

Additionally, the cost of energy purchased from the main grid 7:00 hrs and 18:00 hrs can be expressed as

$$COE = C_f \sum_{t=7}^{18} t_s P_{grid}(t) \quad (8)$$

where $P_{grid}(t)$ denotes the grid power, C_f is the cost of energy per kWh, and t_s is the sampling time.

2.3. Inverter model

Inverter is mandatory in a grid connected photovoltaic system that supplies alternating current (AC) loads. The direct current (DC) electricity which is produced by the photovoltaic array is transformed into alternating current (AC) by the inverter. It is used as a power processing interface between photovoltaic plant and utility grid. Moreover, it provides the synchronization with the utility grid [15, 23]. The power output of the inverter is given by

$$P_{inv}(t) = P_{pv}(t)\eta_{inv}, \quad (9)$$

where $P_{pv}(t)$ is the photovoltaic power output, and η_{inv} denotes efficiency of inverter.

Due to efficient and safe operation, the rating power of inverter should be oversized between 10% and 30% compared to photovoltaic plant [23, 38]. The power rated of inverter is expressed as

$$Pr_{inv} = \sum_{t=7}^{18} P_{pv}(t)X_{pv}K_{inv}, \quad (10)$$

where $P_{pv}(t)$ denotes the photovoltaic power, X_{pv} is the number of photovoltaic panels, while K_{inv} denotes oversize coefficient of inverter.

The lifetime of inverter is less than the lifespan of the project and photovoltaic panels. The number of times during lifetime of project that inverter is required to be replaced is expressed by

$$Rep_{inv} = \frac{N}{L_{inv}} - 1, \quad (11)$$

where N denotes the lifespan of the project, while L_{inv} denotes the lifespan of inverter.

3. Economic analysis

Economic evaluation plays an important role in the optimization design of hybrid energy systems. In this paper, total life cycle cost (TLCC) is considered as economic index to evaluate the project. Total life cycle cost is an economic parameter carried out throughout lifespan of the project by considering the cost of all components. Life cycle cost of each component of the grid connected photovoltaic system is the sum of investment, operation and maintenance cost, replacement cost, and minus salvage cost [36]. The life cycle cost of each component is given by [39]

$$LCC = I_{nvst} + C_{O\&M} + C_{repl} - C_{salv}, \quad (12)$$

where I_{nvst} denotes the investment cost, $C_{O\&M}$ denotes the operating and maintenance cost, C_{repl} is the replacement cost, and C_{salv} denotes the salvage cost.

Investment cost encompasses the cost of purchasing and installing the system. Salvage cost is the incurred cost at the end of the project lifetime. Photovoltaic operation and maintenance cost is assumed to be 1% of the cost of photovoltaic system [16, 40], and installation cost is considered 20% of the initial cost [41]. In this research salvage cost can be neglected because the lifetime of most of components is equal to the lifespan of the project. The present value of operating & maintenance cost is given by [40]

$$\begin{cases} O\&M = (O\&M)_0 \times \frac{1+f}{1-f} \left(1 - \left(\frac{1+f}{1+i} \right)^N \right), i \neq f \\ O\&M = (O\&M)_0 \times N, i = f \end{cases} \quad (13)$$

where, f denotes the inflation rate, i denotes interest rate, and N is the lifespan of the project. In Kenya the inflation rate is 4.65% while the interest rate is 9% [42, 43].

In addition, the total initial investment cost is the sum of each unit price of various components [44]. The total investment cost is given by

$$I_{invst} = IC_{pv} + IC_{inv}, \quad (14)$$

where IC_{pv} denotes the initial cost of photovoltaic, and IC_{inv} denotes the initial cost of inverter. Taking into account 20 % of installation cost, the initial cost of photovoltaic array is given by

$$IC_{pv} = 1.2X_{pv}C_{pv} \sum_{t=7}^{18} P_{pv}(t), \quad (15)$$

where X_{pv} denotes the total number of photovoltaic panels, C_{pv} denotes capital cost of photovoltaic array per KW, and $P_{pv}(t)$ denotes photovoltaic output power.

The initial cost of inverter is given by

$$IC_{inv} = 1.2Pr_{inv}C_{inv} = 1.2X_{pv}K_{inv}C_{inv} \sum_{t=7}^{18} P_{pv}(t), \quad (16)$$

where C_{inv} denotes the capital cost of inverter per kW, Pr_{inv} rating power of inverter, K_{inv} the oversized coefficient of inverter, X_{pv} denotes the number of photovoltaic panels, and $P_{pv}(t)$ denotes photovoltaic power.

The total life cycle cost of the entire system is the sum of LCC of each component, it is given by

$$TLCC = LCC_{pv} + LCC_{inv} \quad (17)$$

where LCC_{pv} denotes the life cycle cost of photovoltaic arrays, and LCC_{inv} denotes the life cycle cost of inverter.

The economic and characteristic parameters of components used to evaluate the total life cycle cost of the entire energy system in this study are illustrated in Table 1. The photovoltaic panel and three phase inverter are the main components which are involved in total life cycle cost.

Table 1: Economic parameters of components [45]

Designations	Photovoltaic Panels	3 Phase Inverter
Cost \$/kW	281	148
(O&M) Cost	1%	-
Instal. cost	20%	20%
Repl. cost	0	148
η_r %	17.5 %	98.5%
Lifespan	25 years	15 years

As shown in Table 1, the three phase DC/AC inverter presents low O&M cost, thus can be neglected. The replacement cost of photovoltaic panel is zero due to the fact that the photovoltaic lifespan is equal at the lifetime of the project.

4. Reliability analysis

The reliability of a distribution power system is geared towards the reducing of frequency or time of interrupted power. This is a great factor to assess system performances. Loss of power supply probability is one of the indexes that is often considered in the evaluation of the reliability of grid connected photovoltaic power system [46, 47].

Generally, loss of power supply probability is a statistical parameter that denotes the deficiency of distributed power system to supply the load demand. The considered site is the university setup where most activities take place from 7:00 hrs to 18:00 hrs. Fortunately, this matches the range of time during which the solar irradiation is available. From the reliability viewpoint, the range of time can then be defined from 7:00 hrs to 18:00 hrs in this study. Assumption is made by neglecting the wired losses within the distribution power system network considered because the available area of installing the photovoltaic panels and the substation are closer to the Engineering workshops. Loss of power supply probability is expressed as [46]

$$LPS P = \frac{\sum_{t=7}^{18} LPS(t)}{\sum_{t=7}^{18} P_L(t)}, \quad (18)$$

where $LPS(t)$ is the loss of power supply at the sampling time t_s , while $P_L(t)$ is the hourly load power demand. Loss of power supply probability is assumed to be between 0 and 1. If the loss of power supply probability equals to 0, it means the load demand is totally covered and if $LPS P = 1$, the load demand is not at all covered by the power supply. The loss of power supply is given by

$$LPS(t) = P_{inv}(t) + P_{grid}(t) - P_L(t), \quad (19)$$

where $P_{inv}(t)$ denotes the power output from the inverter, $P_{grid}(t)$ denotes grid power, and $P_L(t)$ denotes load power demand.

5. Methodology description

In order to carry out the multi-objective optimization sizing of grid connected photovoltaic without energy storage system, the proposed methodology is divided into: multi-objective optimization, objective function and constraints.

5.1. Multi-Objective Optimization

The optimization problem formulated is solved using the MATLAB optimization toolbox which comprises an extensive library of computational algorithms for solving various optimization problems. This optimization under consideration belongs to the category of linear programming minimization problem. As the number of photovoltaic panels must be integer value and the power from the grid utility a continuous value INTLINPROG solver for mixed integer linear programming (MILP) is used.

A mixed integer linear programming is a linear programming (LP) method with the additional restriction that some variables must take integer values. It integrates the branch-and-bound search mechanism, and the cutting plane algorithms through the variation of branch-cut scheme. The technique consists of improving the searching process by systematically enumerating all the

candidates solutions and getting rid of evidently impossible solutions [48, 49, 50, 51, 52].

To solve multi-objective optimization problems, mixed integer linear programming requires scalarization method. Scalarization method transforms multi-objective functions into single aggregating function using weights [53, 54, 55]. Then, the cost function and constraints can be solved using canonical form [56].

In summary, the INTLINPROG solver processes the fundamental strategy to solve mixed integer linear programming as follows

1. Restrict the size of the problem by using linear program preprocessing.
2. Determine an initial released linear programming.
3. Execute mixed integer program preprocessing to restrain the linear program relaxing of the mixed-integer problem.
4. Determine cut generation to further consolidate the linear programming relaxing of the mixed integer problem.
5. Retrieve integer-feasible solutions heuristics.
6. Apply the branch-and-bound algorithm to seek the optimal solution.

INTLINPROG function is free to set options for branching rules and solves the problem at any of these enumerated stages.

5.2. Objective function

The total life cycle cost of the project and energy purchased from the utility grid are the objective functions which are considered to be minimized. The objective function can be expressed as

$$\min F = \min \left(w \sum_{q=1}^p \left(I_{nvst}(q) + C_{O\&M}(q) + C_{repl}(q) - C_{salv}(q) \right) + (1 - w) C_f \sum_{t=7}^{18} t_s P_{grid}(t) \right) \quad (20)$$

where p is the number of components to be purchased related to the project, I_{nvst} denotes the investment cost, $C_{O\&M}$ denotes the investment cost, C_{repl} denotes the replacement cost, C_{salv} is the salvage cost, C_f is the cost of energy per kWh, t_s is the sampling period, $P_{grid}(t)$ denotes grid power, w denotes the weighing factor selected to point out the relative importance of lowering each term in the objective function. Owing to the hourly solar irradiation and power demand data, the sampling period of 1 hour is chosen in this study.

The optimization variables are the number of photovoltaic panels and the hourly grid powers. The number of photovoltaic arrays must be an integer value, while the hourly powers from utility grid are continuous variables. Therefore, taking into account the decision variables, the objective function Eq. 20 can be rewritten as

$$\min F = \min \left(w \left[C_{pv} \left(1.2 + 0.1 \frac{1+f}{1-f} \left(1 - \left(\frac{1+f}{1+i} \right)^N \right) \right) + 2.2 \eta_{inv} K_{inv} C_{inv} \sum_{t=7}^{18} t_s X_{pv} P_{pv}(t) P_{r_{pv}} \right] + (1-w) C_f \sum_{t=7}^{18} t_s P_{grid}(t) \right) \quad (21)$$

where C_{pv} denotes the capital cost of photovoltaic arrays pr kW, C_{inv} denotes the capital cost of inverter, f denotes the inflation rate, i denotes the interest rate, η_{inv} denotes the efficient of inverter, K_{inv} denotes oversize coefficient of inverter, t_s the sampling period, X_{pv} denotes the number of photovoltaic arrays, and Pr_{pv} denotes the power rated of photovoltaic, N is the lifespan of the project, w is the weighting factor, C_f is the cost of energy per kWh, $P_{grid}(t)$ denotes power grid.

The first term of the objective function represents the total life cycle cost of the project, and second term represents energy purchased from the grid. In this research, the priority of total life cycle cost and energy purchased is assumed to be balanced. Thus, the weighing factor has been selected to 0.5.

5.3. Constraints

The entire grid connected photovoltaic system is subject to some operational and technical constraints between power generations and load demand. In order to maximize the reliability, the LPSP(t) is given by

$$\begin{cases} LPS P \geq 0 \\ \sum_{t=7}^{18} X_{pv} P_{inv}(t) + P_{grid}(t) - P_L(t) \geq 0 \\ \sum_{t=7}^{18} X_{pv} P_{inv}(t) + P_{grid}(t) \geq \sum_{t=7}^{18} P_L(t) \\ -\left(\sum_{t=7}^{18} X_{pv} P_{inv}(t) + P_{grid}(t)\right) \leq -\sum_{t=7}^{18} P_L(t) \end{cases} \quad (22)$$

The objective function and constraints are solved using the canonical form in [56], such that

$$\min f^T X \quad (23)$$

subject to

$$\begin{cases} AX \leq b \\ L_b \leq X \leq U_b \\ X = [X_{pv}, \sum_{t=7}^{18} P_{grid}(t)]^T \\ X_{pv} \in \mathbf{N} \\ P_{grid}(t) \in \mathbf{R} \end{cases} \quad (24)$$

The $f^T X$ denotes the objective function to be minimized, X consists of the decision variables that is a vector representing the optimal number of photovoltaic panels and the energy purchased from the grid, A denotes matrix associated with inequality constraints, and b is the scalar related to inequality constraint; L_b is the lower bound vector, and U_b is the upper bound vector of the decision variables.

From Eq. 21, the objective function f^T is expressed as

$$f^T = [541.3277w, (1 - w)C_f] \quad (25)$$

The vector X consists of all the decision variables, it is expressed as

$$X = \left[X_{pv}, \sum_{t=7}^{18} P_{grid}(t) \right]^T \quad (26)$$

From the reliability constraint measured by the loss of power supply probability Eq. 22, inequality constraint AX is expressed as

$$AX = \sum_{t=7}^{18} -X_{pv}P_{inv}(t) - P_{grid}(t) \quad (27)$$

where $P_{inv}(t)$ is the output power of inverter at time t , $P_{grid}(t)$ is the power supplied by the utility grid. From Eq. 7 and Eq. 27 the matrix A is given by

$$A = \left[\sum_{t=7}^{18} -P_{inv}(t), -1 \right] \quad (28)$$

The scalar b is given by

$$b = \sum_{t=7}^{18} -P_L(t) \quad (29)$$

As defined in section 2, the lower (L_B) and upper (U_B) bounds of the decision variables given in canonical form are expressed as

$$L_B = [0, 0] \quad (30)$$

$$U_B = [459, +\infty] \quad (31)$$

6. Case study

The Engineering workshops at Jomo Kenyatta University of Agriculture and Technology is the case used in this study. It is located in Kenya, Kiambu county, Juja town at -1.099° latitude and 37.014° longitude. Fig. 2 depicts the localization of Jomo Kenyatta University of Agriculture and Technology [57, 58, 59].

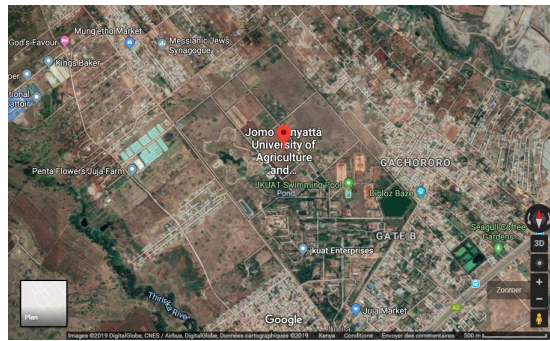


Figure 2: Localization of Jomo Kenyatta University of Agriculture and Technology

The setup is the university workshops where most activities happen from 7:00 hrs to 18:00 hrs. The facility consists of various workshops such as: machine tools, carpentry, Iplic, foundry, welding, plumbing, and fitting workshops. Each workshop has various three and single phase

types of electrical loads. Fig. 3 shows daily power consumption of Engineering workshops.

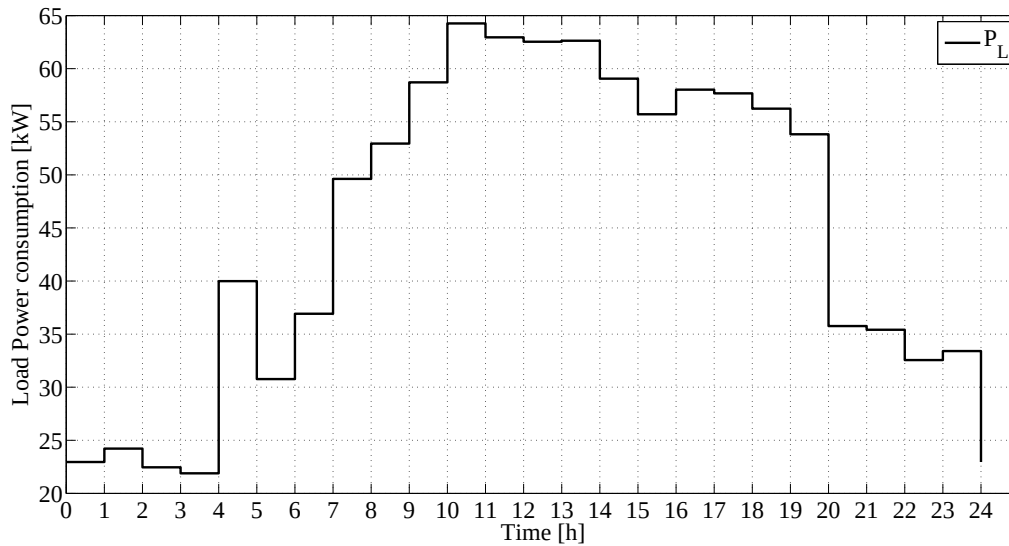


Figure 3: Daily power demand of Engineering workshops at JKUAT

The daily power consumption corresponds to the operation of several type of equipments such as machine tools, welding machines, and others. The daily load profile is mainly represented by a suite of power demand. The power demand is taken as constant over each time-step of 1 hour. Systematically, the sampling period is 1 hour as illustrated in Fig. 3.

The daily profile load has been determined by using three phase power logger Fluke 1735. Its sampling time was set up to 1 minute and the hourly peak power has been selected. The logging has been done for several days during the peak period of consumption. It can be observed from the dynamic variation of the power load demand that during night hours the power consumption is low. It starts increasing early in the morning. From the daily load profile, it is shown that the peak load is 64.26 kW between 10:00 hrs and 11:00 hrs and the total daily energy consumption is 1090.5 kWh/day while daytime energy demand of 700.3510 kWh can be deduced. The selected daytime is from 7:00 hrs to 18:00 hrs when the solar irradiation is available at the site. The workshops power system supply is currently provided by the national grid and diesel generator as backup system. A diesel generator of 250 kVA is used for uninterrupted power supply whenever there is outage of power from main grid. Owing to high levelized cost of energy of diesel generator, the assumption is made by considering the availability of the national grid over 24 hours. Thus, the grid utility will be the unique source to feed the load demand when solar radiation is not available. The hourly solar radiance and ambient temperature data of the site are obtained from [60]. The optimal sizing of the grid connected photovoltaic system is derived from a selected day of the most unfavorable month in terms of irradiation. In Kenya, July is considered as the worst month due to the fact the monthly average irradiation is the lowest in comparison with other months. July

average irradiation is estimated to 4.4 kWh/m²/day [61]. Figure 4 depicts the solar irradiation of the selected day.

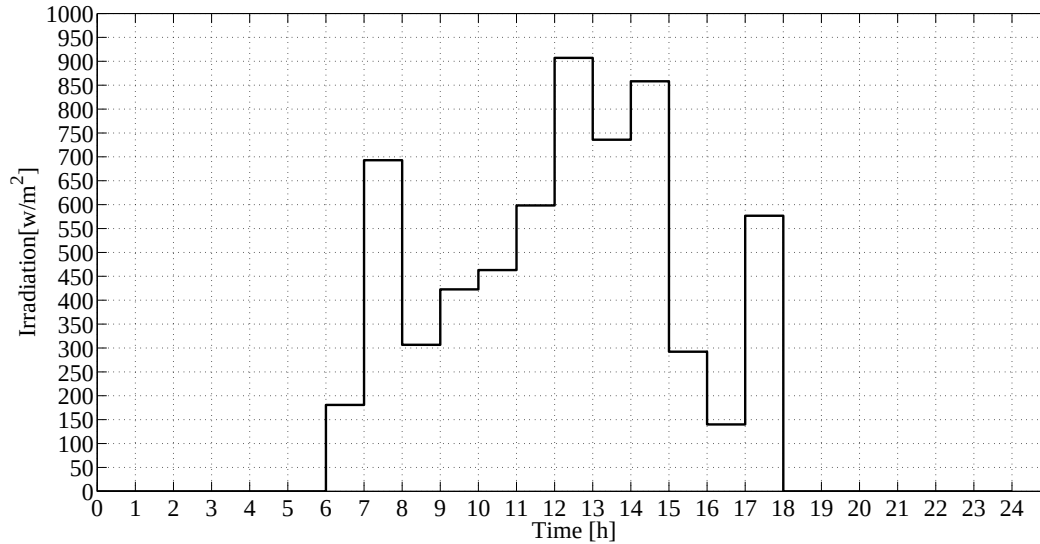


Figure 4: Solar irradiation of the selected day

Apart from economic parameters displayed in section 3, Table 2 shows other different constant parameters used in this study.

Table 2: Various parameters used in this study

Parameters	Designations	Values
Project lifetime	N	25 years
Lifetime of PV		25 years
Rated power of PV	P_r	340 W
Rated Efficiency of PV	η_r	17.5%
Area of PV	A_{pv}	1.96 m ²
Available Area	A_T	900 m ²
Inverter lifetime	L_{inv}	15 years
Inverter Efficiency	η_{inv}	98.5%
Inverter factor	K_{inv}	1.25
Sampling time	t_s	1 h
Inflation rate	f	4.65%
Interest rate	i	9%
weighing factor	w	0.5
Lower bound	L_B	[0 0]
Upper bound	U_b	[459 +∞]
Cost of energy	C_f	0.12 \$/kWh

7. Results and discussions

This section presents the results obtained for multi-objective optimization sizing of grid connected photovoltaic without battery system. The baseline of Engineering workshop is the national grid with diesel generator backup system. An assumption is made by considering the availability of grid utility over 24 hours. Hence, from the sizing perspective, the baseline is simply the utility grid in this study.

The results discussed are related to the grid connected photovoltaic system without the diesel generator due to its high levelized cost of energy. The objective function has been solved using INTLINPROG solver for mixed integer linear programming problem as discussed in section 5. The photovoltaic power plant has been designed in order to supply the load demand during the daytime hours from 7:00 hrs to 18:00 hrs while maximizing the reliability requirements. Due to the unavailability of radiance solar for the remaining times, grid utility will supply the loads power demand at those specific times.

As JKUAT does not have a power purchase agreement (PPA) with Kenya power lighting company (KPLC), the entire energy system has been designed such that the losses that would arise by feeding power to the grid are minimized. Table 3 displays the values of decision variables and total life cycle cost of project when the weighting factor is equal to 0.5.

Table 3: Simulation results obtained

Parameters	Designation	Values
Number of PV panels	X_{pv}	354
Energy purchased	E_{grid}	0.6022 kWh
Total life cycle cost	$TLCC$	191630\$
Daily energy saving		64.16%
Revenue		574596.25\$

The same results of Table 3 arise also when a value within the range of within the range of $0.376 \leq w \leq 0.664$ is assigned to the weighing factor. It can be deduced that the optimal number of photovoltaic panels is 354, the optimal energy from the grid in daytime is 0.6022 kWh, while the total life cycle cost is found to be 191630\$. With this energy system, the grid energy during the daytime between 7:00 hrs and 18:00 hrs has considerably been lowered from 700.3510 kWh to 0.6022 kWh.

The daily energy to supply the loads demand of the Engineering workshop is approximately 1090.5 kWh. Considering the baseline and the cost of electricity 0.12 \$/kWh in Kenya [62], the daily energy purchased from the grid cost 130.86\$. The total energy generated by the photovoltaic panels is 699.749 kWh. The economical benefit consists in a saving of 83.97\$ per day in comparison with the baseline considered in this study. The entire energy system can daily yield energy saving up to 64.16 %.

Varying the weighting factor between $0 \leq w \leq 0.375$ the photovoltaic power plant is prioritized as apposed to the grid utility. The photovoltaic power plant system is oversized so that the optimal energy of the grid utility starts becoming negative. This means that the grid utility absorbs more electrical energy from the photovoltaic power plant. As stated, there is no need to oversize the photovoltaic plant because the facility does not have power purchased agreement with Kenya power and lighting company. This can bring losses that occur by feeding the excess photovoltaic energy to the utility grid. On the other hand, by assigning the weighting factor between $0.665 \leq w \leq 1$ all the power that feeds the loads comes from the grid utility. The INTLINPROG solver has selected the lower bound of the number of photovoltaic panels which is zero as defined. This means the grid is more prioritized to cover the loads demand in the sizing process than the photovoltaic plant.

Besides, the total life cycle cost of the project has been estimated for 25 years, photovoltaic energy generated during this lifetime of the project can be evaluated to be 776226.25\$. Therefore, the savings which can be generated from installing the photovoltaic energy system connected to the grid is approximately 574596.25\$. This shows that the grid connected photovoltaic batteryless system can clearly be profitable for end-user customer at demand side in this region for the chosen facility.

8. Conclusion

This paper develops a novel optimization methodology based on multi-objective optimal sizing of grid tied photovoltaic without energy storage systems. The optimization design seeks to lower the total life cycle cost and energy purchased from the grid while at the same time satisfying the reliability constraint. In this study, the reliability has been measured by the loss of power supply probability index. Mixed integer linear programming has been used to find the optimal values of decision variables, namely the number of photovoltaic panels and energy purchased from the utility grid. This model considerably lowers the daily electricity cost by reducing the energy purchased during daytime which is much beneficial at customer side. The approach ensures efficient and reliable grid connected photovoltaic system and can be suitable for any application wherever energy storage systems may present prohibitive price. Systematically, a case study has been considered in order to supply the load of Engineering workshops at Jomo Kenyatta University of Agriculture and Technology. From the optimization results obtained, the optimal sizing consists of 354 photovoltaic panels, the energy purchased from the grid during the daytime is 0.6022 kWh, and the total life cycle cost of the project is estimated 191630 \$. With these optimal values, the proposed power supply cover the load demand during daytime hours from 7:00 hrs to 18:00 hrs while guaranteeing the reliability requirement. Grid utility can be the main power source to supply load demand only during the remaining times. The daily energy saving is up to 64.16% in comparison with the baseline, and the economical benefit for the lifetime of the project arises to 574596.25\$. Varying the weighing factor affects the multi-objective optimization sizing of grid connected photovoltaic accordingly. The developed multi-objective optimal sizing technique can be applied to other areas or locations although it has been carried out with particular dataset of

Engineering workshops at Jomo Kenyatta University of Agriculture and Technology as case study. In other words, this optimization design can be used to find out the optimal sizing of grid tied photovoltaic without storage energy systems at any location aiming at lowering the total life cycle cost and energy purchased whilst maximizing the reliability.

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- Mixed integer linear programming is used to determine the decision variables.
- The optimal number of photovoltaic panels is 354.
- The total life cycle cost is 191630 \$ and daytime grid energy is 0.622 kWh.
- The daily potential energy saving is up to 64.16%.